PART I: FOUNDATIONS

Topic 2: Time Series vs. Cross-Sectional Data

The Core Concept

- **Cross-Sectional Data:** A "photograph" — a snapshot of many subjects at one point in time.
- **Time Series Data:** A "movie" — a recording of one subject across many points in time.

Why It Matters

If we only compare precincts at one moment in time (cross-sectional), we might conclude that Precinct South is "more dangerous" because its numbers are higher. But tracking each precinct over time (time series) reveals a very different story.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

- **Cross-Sectional View (December only):** North = 11, South = 26. Conclusion: "South is worse."
- **Time Series View (full year):** South stays between 22–28 all year — stable. North doubles from 12 to 25 — volatile.

💡 **The Insight:** Cross-sectional data helps answer "where to allocate resources today." Time series data answers "is a specific unit's situation improving or deteriorating?"

Key Takeaway: Use cross-sectional data for resource allocation; use time series data for policy evaluation.